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**Machine Learning-Based Quantification of Ovarian Follicle Stages in *Heterocephalus*
*glaber***

Julian Coles, Martin Orkuma, Pamela Styborski, and Silvia Tenempaguay-Nunez
New College of Interdisciplinary Arts and Sciences,
Arizona State University, Tempe, Arizona
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Abstract

Ovarian histopathology plays a critical role in understanding follicle development and disease progression, yet manual identification and quantification of ovarian follicles is a labor-intensive process that limits large-scale histological research. The naked mole rat (*Heterocephalus glaber*) is a unique model for studying reproductive longevity and resilience compared to mammals and other rodents, yet automated tools for analyzing its ovarian architecture are lacking. This study presents a **machine-learning pipeline for tile-level classification** of ovarian follicle stages in *H. glaber* histology images. Using whole-slide images from the Multispecies Ovary Tissue Histology Electronic Repository (MOTHER), and corresponding metadata from the MOTHER dataset, we implement a deep learning pipeline based on a ResNet34 convolutional neural network (CNN). QuPath-generated annotations were used to train a **ResNet34 CNN classifier** to assign follicle-stage labels to image tiles, enabling **WSI-level aggregation of tile-level predictions**. The trained model is evaluated for its ability to accurately detect and count follicle types, with the broader goal of improving reproducibility and scalability in ovarian histological analysis. This work highlights the potential of integrating AI-driven image analysis with curated histopathology datasets to advance studies of ovarian biology, while also addressing current limitations in data variability and annotation quality.

Summary Sentence: An automated machine learning pipeline enables reproducible quantification of ovarian follicle stages in the naked mole rat (*Heterocephalus glaber*).

Keywords: naked mole rat, *Heterocephalus glaber*, machine learning, ovarian follicles, digital pathology, MOTHER database, convolutional neural network (CNN).

Introduction

Histology has traditionally relied on expert visual interpretation of tissue sections, but the rise of digital slide imaging and advances in high-performance computing have brought machine-learning (ML) into the field with promising impact. Interactive digital pathology platforms, such as QuPath and ImageJ, are increasingly being used to help analyze tissue images by automating time-consuming tasks, measuring complex features, and detecting subtle patterns that may be difficult to recognize consistently by eye.

Deep learning methods, especially convolutional neural networks, have shown strong performance in areas such as tumor detection, tissue classification, grading, and even predicting molecular or clinical outcomes directly from histologic images [1]. However, one of the limitations observed in the general application of machine-learning in histology is reproducibility and data quality. These concerns directly affect whether ML tools can be trusted beyond controlled research environments, particularly when it comes to large, variable datasets.

The Multispecies Ovary Tissue Histology Electronic Repository (MOTHER) offers a rich resource of digitized ovarian histology images and detailed metadata that align with FAIR (Findable, Accessible, Interoperable, Reusable) principles, enabling cross-species comparisons and reuse of valuable tissue data [2]. The challenge of reproducibility and data quality in the application of ML is particularly relevant for MOTHER, which includes these diverse, multi-species ovarian tissue slides and associated XML metadata, increasing both the complexity and potential of the dataset. To mitigate some of these concerns, a more targeted approach to this dataset is proposed to combine identification, segmentation, and counting within a unified analytical pipeline applied to an underrepresented species, *Heterocephalus glaber*. *H. glaber*, more commonly known as the naked mole rat, serves as a unique model in the study of fertility

and ovarian health, compared to the typical decline in mammalian reproductive health seen in a natural lifecycle [3].

Machine learning methods can support time-intensive tasks such as follicle-stage classification and detection of atresia or structurally abnormal tissue regions, enabling scalable and reproducible analysis across large image collections. This reduces observer variability and allows researchers to extract consistent, quantitative features that can be compared across species, age groups, and experimental conditions [4]. By prioritizing generalizability and integrating image analysis with structured metadata, the application of ML to the MOTHER dataset can move beyond exploration toward reliable, scalable tools that support reproducible ovarian histology research across species.

Materials and Methods

Workflow Pipeline

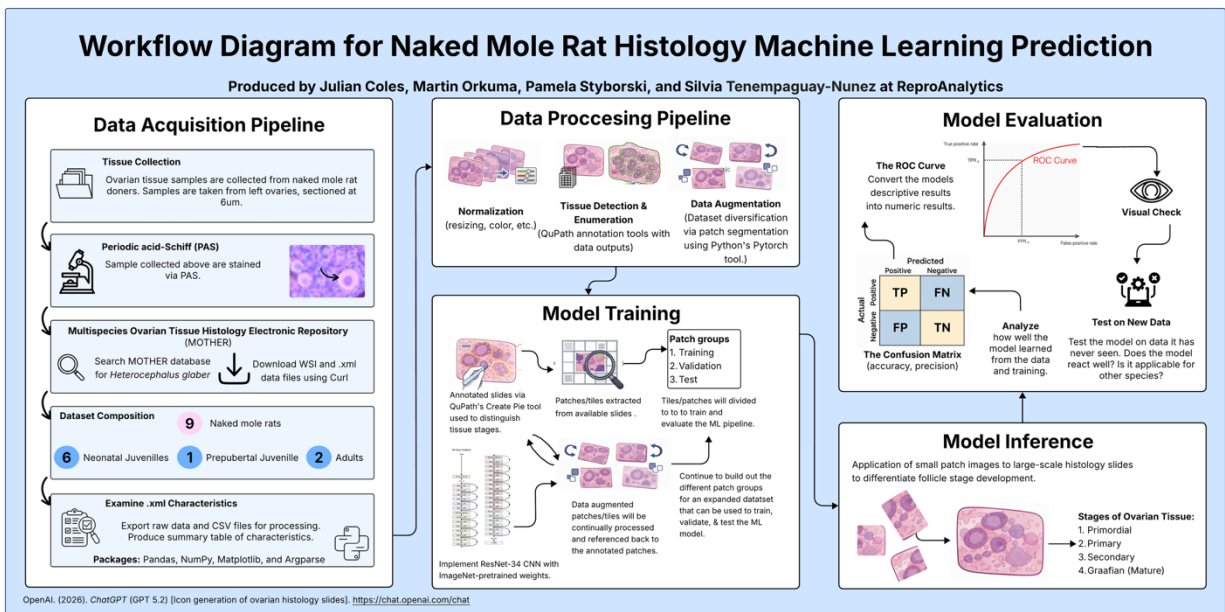


Figure 1. Automated Workflow Pipeline

Data Source and Acquisition

Digital histology slides and structured metadata were sourced from the Multispecies Ovary Tissue Histology Electronic Repository (MOTHER), with funding from the National Science Foundation (NSF DBI-2054061, 2021 – 2025) [2]. The dataset comprises nine individual naked mole rat donors (image snapshots included in Supplemental Figure 4), representing diverse life stages: six neonatal juveniles (aged 1, 5, 8, 15, 28, and 90 days), one prepubertal juvenile (180 days), and two adults (3 years). All samples were derived from left ovary tissue, sectioned at a thickness of 6 μm , and stained with periodic acid-Schiff (PAS).

	Source	Common Name	Scientific Name	Donor ID	Donor Life Stage	Donor Age (Years)	Donor Age (Days)	Donor Sex	Ovary Sampled
0	MDB0000530-P28-Ova-106_s55.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P28-Ova-106	neonatal	0	28	female	left
1	MDB0000531-P1-11-s40.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P1-Ova11	neonatal	0	1	female	left
2	MDB0000532-P15_F09OV-s83.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P15-F09	neonatal	0	15	female	left
3	MDB0000533-P180_6Mon-6b-s86.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_NMR-6Mon-6B	prepubertal	0	180	female	left
4	MDB0000534-P5-23-s13.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P5-Ova23	neonatal	0	5	female	left
5	MDB0000535-P8-F02-OV-s135.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P8-F02	neonatal	0	8	female	left
6	MDB0000536-P90-NMR_3Mo-3B-s96.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_P90-3B	neonatal	0	90	female	left
7	MDB0000537-2653-3y-Exsub-s19.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_2653	adult	3	0	female	left
8	MDB0000538-2659-3y-Sub-s109.xml	naked mole-rat	Heterocephalus glaber	MABE_HG_2659	adult	3	0	female	left

Table 1. Summary of XML derived characteristics of the selected samples.

Data Preprocessing/Exploratory Data Analysis (EDA)

We conducted a technical validation of the MOTHER dataset using Python (v3.11) to ensure model robustness, profile the XML metadata, and characterize dataset composition. Visual quality control was conducted using PNG slide previews. EDA objectives included characterizing composition of the dataset, examining image-level characteristics, and generating

initial image analysis. Image-level analysis confirmed uniform magnification (40x) and staining protocols (PAS) across all samples.

Tissue-specific RGB histograms were generated by masking near-white background pixels to analyze staining intensity (Figure 1). Color distributions revealed that red-channel intensity was particularly informative for PAS-positive basement membranes, whereas blue-channel intensity reflected the density of nuclear counterstain. In a summary of the statistics of this RGB channel analysis, both the neonatal and adult samples are captured within the standard deviation of each individual color channel between the respective samples in each previously identified donor age group (Supplemental Table 2). There are no significant outliers; however, the spread of each histogram is considerably wide.

Image Analysis and Annotation

We employed a kernel density estimation (KDE) plot to compare the nuclear area distributions across samples of total nuclear counts (Supplemental Figure 7). The KDE estimates a continuous probability density to compare the shape of the curves independently. Each curve on the graph represents a separate slide represented by a nuclei measurement ID that coincides with the accession numbers, in ascending order, located in the Subject ID.

A layered plot allows for a visualization of the full distribution of each sample to assess the alignment and overall shape of each distribution for comparison. This approach also allows for observations made on outliers in both peak height and potential anomalies in skew. In this case, we have some skewness observed across the dataset. The distribution of each slide was normalized independently to compare shapes; the curves are observed almost on top of each other which might indicate nuclear size variation within and across samples. In another output

provided by the QuPath annotation analysis, similar trends are seen in the boxplots of area distribution across all nine samples (Supplemental Figure 2).

Whole-slide images in .tiff format were analyzed using QuPath (v0.6.0). QuPath was selected as the primary annotation and exploratory analysis tool because it provides native support for whole-slide histology images, hierarchical region-of-interest annotation, and structured export formats compatible with downstream machine learning workflows. It is also the tool of choice for whole-slide image analysis in the MOTHER database [2]. To establish a ground-truth dataset, three of the nine slides were selected for manual annotation. QuPath was used to perform structured image annotations on the three whole-slide TIFF images following quality control and metadata organization.

Automated tissue detection was first applied to identify and isolate relevant tissue regions. A supervised pixel classifier was then trained to distinguish key histological components, enabling accurate segmentation of tissue compartments. These classifications were used to generate object-level annotations that preserved spatial relationships across the slide. Annotation of ovarian follicle classes, including germ cells, primordial, transitional primordial, primary, secondary, multilayer, antral, and atretic antral follicles, as well as stromal regions, was guided by a standardized key derived from supplemental instructional materials available through Indiana University Scholar Works [8]. This reference framework improved consistency across annotations and reduced inter-operator variability by aligning labeling decisions with established morphological criteria.

For tile generation, QuPath's scripting interface was used to enable reproducible and scalable extraction of image patches from annotated regions. Using functionality described in the QuPath image export documentation and implemented within the script editor, tiles were

programmatically generated with fixed dimensions and resolution, with optional overlap to retain contextual continuity between adjacent regions [9]. The resulting high-resolution tiles preserved fine morphological detail and provided a well-structured dataset suitable for downstream computational analysis in our machine learning model development.

Model Architecture

The annotated output data from QuPath were used as input for the model training pipeline. Ovarian follicle-stage classification was performed using a **ResNet34 CNN classifier** pretrained on the ImageNet dataset and implemented with the fastai library in Python [7]. The model was trained on 385 image tiles to generate **tile-level predictions** across eight tissue classes: primordial, transitional primordial, primary, multilayer, antral, atretic antral, stroma, and germ cells.

To prevent data leakage, where tiles from the same donor appear in both training and evaluation sets, we employed data splitting at the accession level rather than the individual tile level. We accomplished this by saving images in different directories. Accessions were randomly assigned to training (70%), validation (15%), and test (15%) partitions, ensuring that all tiles derived from a given whole-slide image were confined to a single split.

Data Augmentation

To improve generalization, we added extensive data augmentation to training tiles at each epoch. Augmentations included vertical and horizontal flips, random rotations up to 90°, mild zoom (up to 1.2×), brightness and contrast jitter to simulate staining variation across slides, and mild perspective warping. These are all safe transformations for histology images because tissue

sections have no canonical orientation, and staining intensity naturally varies between slides. The goal is to effectively multiply the training set by exposing the model to many different-looking variants of each tile.

Class Imbalance Handling

We noticed a substantial class imbalance among follicle types in ovarian histology images. For instance, stromal and background regions dominate, and follicle classes such as antral or atretic antral are comparatively rare. To mitigate this imbalance, we applied inverse-frequency class weighting to the cross-entropy loss function. For each class, the weight was computed as the reciprocal of the number of training tiles in that class, then normalized such that the weights summed to the total number of classes. This ensured that misclassifications of underrepresented follicle stages incurred proportionally greater loss during optimization. The effect of this intervention on prediction distribution is reported in the Results (Table 2).

Training Procedure

Training was conducted in two phases, frozen and unfrozen phases. In the first phase, all pretrained convolutional layers were frozen and only the newly initialized classification head was trained for up to 5 epochs at a learning rate of 1×10^{-3} . In the second phase, all layers were unfrozen and the full network was trained for up to 10 additional epochs using discriminative learning rates, with a one-cycle learning rate ranging from 1×10^{-5} in the earliest layers to 1×10^{-3} in the classification head.

This graduated approach allows the deeper, more general feature extractors to update conservatively while permitting the task-specific layers to adapt more aggressively. A batch size

of 16 was used throughout. Early stopping with a patience of 3 epochs monitored validation loss in both phases, and the model checkpoint with the lowest validation loss was retained for evaluation and inference.

Evaluation

Model performance was assessed on the held-out test partition using a confusion matrix, per-class receiver operating characteristic (ROC) curves with area under the curve (AUC) scores, and a top-losses visualization highlighting the five most confidently incorrect predictions to support qualitative error analysis. Tile-level predictions and associated confidence scores were recorded for downstream WSI-level prediction aggregation.

Whole-Slide Inference

We included a post-training step; the trained model was applied to full whole-slide images from the MOTHER database. Slides were read at a reduced resolution level (level 1) using the tiff file Python library, tiled into 224×224-pixel patches with a stride of 224 pixels, and filtered for tissue content using a mean-intensity threshold of 220. Each tissue-containing tile was classified by the ResNet34 classifier, and tile-level predictions were aggregated into per-slide class distributions as well as spatial prediction maps recording the class assignment and tile coordinates for each patch.

Results

Introduction

The results show the feasibility of applying supervised machine learning to tile-level classification and WSI-level prediction aggregation of ovarian follicle stages in naked mole-rat histology images. Using nine whole-slide images from the MOTHER database and manually annotated tile data through QuPath, we evaluated training set composition, class imbalance, model behavior during training, classification performance on a held-out test set, and whole-slide inference across donors representing neonatal, prepubertal, and adult life stages.

In addition to measuring predictive performance through validation loss, error rate, confusion matrix, and ROC analysis, we examined how model predictions were distributed across full slides to identify class-specific patterns associated with developmental stage. Collectively, these results provide an initial assessment of both the strengths and limitations of the pipeline and support its use as a reproducible framework for automated ovarian histology analysis in *Heterocephalus glaber*.

Training Data Composition

The training dataset comprised 385 tiles across eight follicle classification categories, manually annotated from QuPath exports. A total of 84 tiles were held out for the test set and 84 for validation, with splits performed at the accession level to prevent data leakage between slides from the same specimen.

Class distribution of follicles was quite imbalanced (Figure 2). Stroma accounted for the largest share with 118 training tiles, followed by primordial (73 tiles) and antral (46 tiles). The most underrepresented classes were primary and transitional primordial, each with only 22

training tiles. To mitigate this imbalance, inverse-frequency class weights were applied to the cross-entropy loss function, assigning weights of 1.75 to the minority classes and 0.33 to stroma.

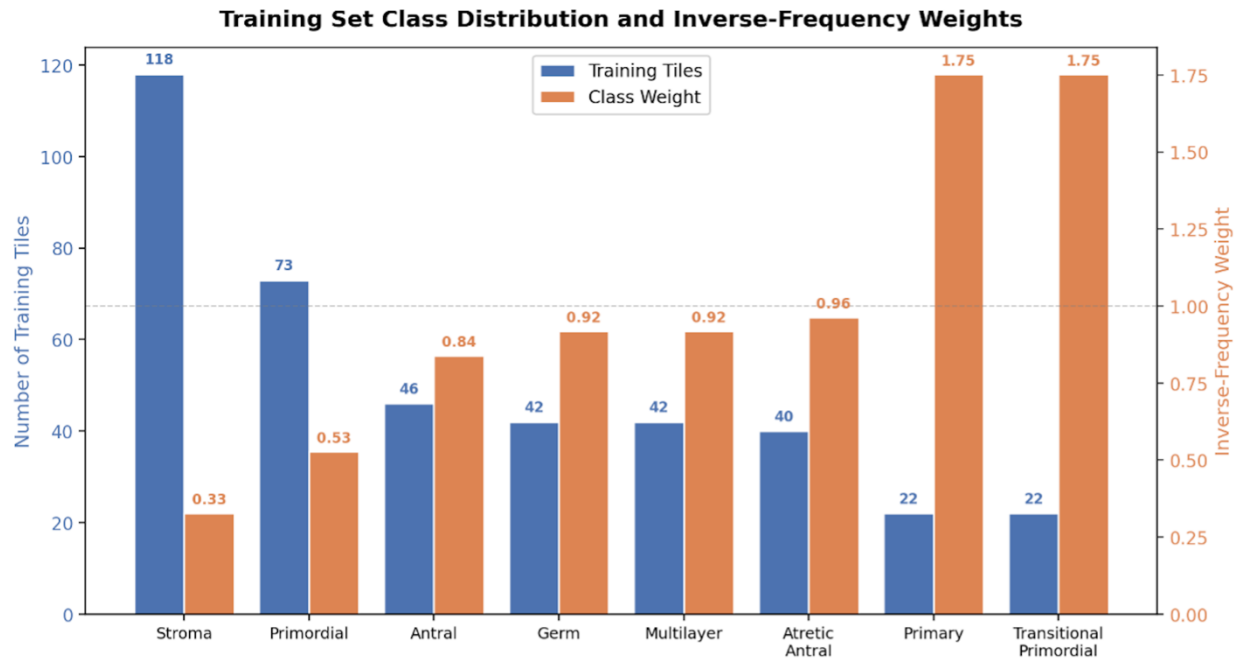


Figure 2: Training set class distribution (blue bars) and corresponding inverse-frequency weights (orange bars).

Table 2 compares the number of test tiles predicted for each class before and after the application of weighted-loss and augmentation. Prior to these interventions, the model heavily favored the majority of classes, predicting stroma for 24 of 84 test tiles. After applying inverse-frequency weighting and augmentation, the prediction distribution became more balanced, with

stroma reduced to 12 predictions and underrepresented classes such as primary increasing from 2 to 11 predictions.

Class	Before	After
Stroma	24	12
Primordial	18	14
Primary	2	11
Antral	11	17
Atretic Antral	7	8
Germ	11	14
Multilayer	6	1
Transitional Primordial	5	7

Table 2. Test prediction counts before and after applying weighted-loss and augmentation (84 tiles total).

Model Training

The pipeline used ResNet34 pretrained on ImageNet, trained with the fastai library in two phases on a CUDA-enabled GPU. All tiles were resized to 224×224 pixels and normalized to ImageNet statistics. Data augmentation included vertical and horizontal flips, 90-degree

rotations, zoom (up to 1.2×), brightness and contrast jitter (up to 0.3), and perspective warping (up to 0.15), applied with 75% probability per transform.

Phase 1 (Frozen Feature Extraction). Only the classification head was trained for 5 epochs at a learning rate of 0.001. Validation loss decreased monotonically from 1.43 to 0.92, and error rate dropped from 46.4% to 32.1%. A new best model checkpoint was saved at each epoch, indicating consistent improvement throughout the phase. The best model was found at Epoch 3 with a validation loss of 0.785 and an error rate of 26.2%. Training times remained consistent across epochs, averaging approximately 3 minutes and 18 seconds per epoch.

Phase 2 (Unfrozen Fine-Tuning). All layers were unfrozen and trained with discriminative learning rates (1×10^{-5} to 1×10^{-3}) using a one-cycle policy. The model found a marginal improvement at epoch 0 (valid loss = 0.783), but validation loss increased at epochs 1 (1.12), 2 (1.12), and 3 (1.17), triggering early stopping with patience = 3. The best checkpoint from the end of Phase 1 (valid loss = 0.783) was restored.

Test Set Evaluation

The best model checkpoint was evaluated on 84 test tiles. The model achieved an overall error rate of 32.1% on the test set, indicating moderate performance when applied to previously unseen image data. Figure A1 (in the appendix) presents the distribution of predicted classes. Primordial and stroma received the most predictions (17 tiles each), consistent with their higher representation in training. Atretic antral was predicted for 15 tiles, while antral received the fewest predictions (5 tiles). The distribution of predictions suggests that the model more readily

identified certain dominant or morphologically distinct classes, whereas other categories were represented less consistently across the test set.

Confusion Matrix

A confusion matrix was constructed to evaluate per-class classification performance of the ResNet34 model across eight tissue classes (Figure 3). The highest diagonal values were observed for primordial (11 of 15 correctly classified), germ (7 of 8), antral (6 of 10), and transitional primordial (5 of 5). Primary achieved 4 of 5 correct classifications. Stroma had 14 of 24 tiles correctly classified, with 5 tiles misclassified as primordial and 3 as germ. Atretic antral achieved 4 of 8 correct classifications, with the remaining tiles distributed across antral (1) and multilayer (3). Multilayer was the most frequently misclassified class, with only 3 of 9 tiles correctly classified; 3 tiles were misclassified as primordial and 2 as atretic antral.

		Confusion matrix							
Actual	antral	6	3	1	0	0	0	0	0
	atretic antral	1	4	0	3	0	0	0	0
	germ	0	1	7	0	0	0	0	0
	multilayer	1	2	0	3	0	3	0	0
	primary	0	0	0	0	4	0	1	0
	primordial	0	0	0	1	1	11	2	0
	stroma	0	0	3	0	1	5	14	1
	transitional primordial	0	0	0	0	0	0	0	5
		antral	atretic antral	germ	multilayer	primary	primordial	stroma	transitional primordial
		Predicted							

Figure 3. Confusion Matrix of Predicted vs. Actual Cell Types

ROC Curves

A receiver operating characteristic (ROC) curve was generated to evaluate discriminative performance across all eight classes (Figure 4). The graph shows the true positivity rate of the model compared to the false positive rate. Per-class area under the curve (AUC) scores were as follows: transitional primordial (AUC = 1.00), primary (AUC = 0.99), germ (AUC = 0.98), antral (AUC = 0.95), primordial (AUC = 0.93), stroma (AUC = 0.91), multilayer (AUC = 0.86), and atretic antral (AUC = 0.84). All classes achieved AUC values substantially above the random-chance baseline of 0.50, with a mean AUC of 0.93 across all classes.

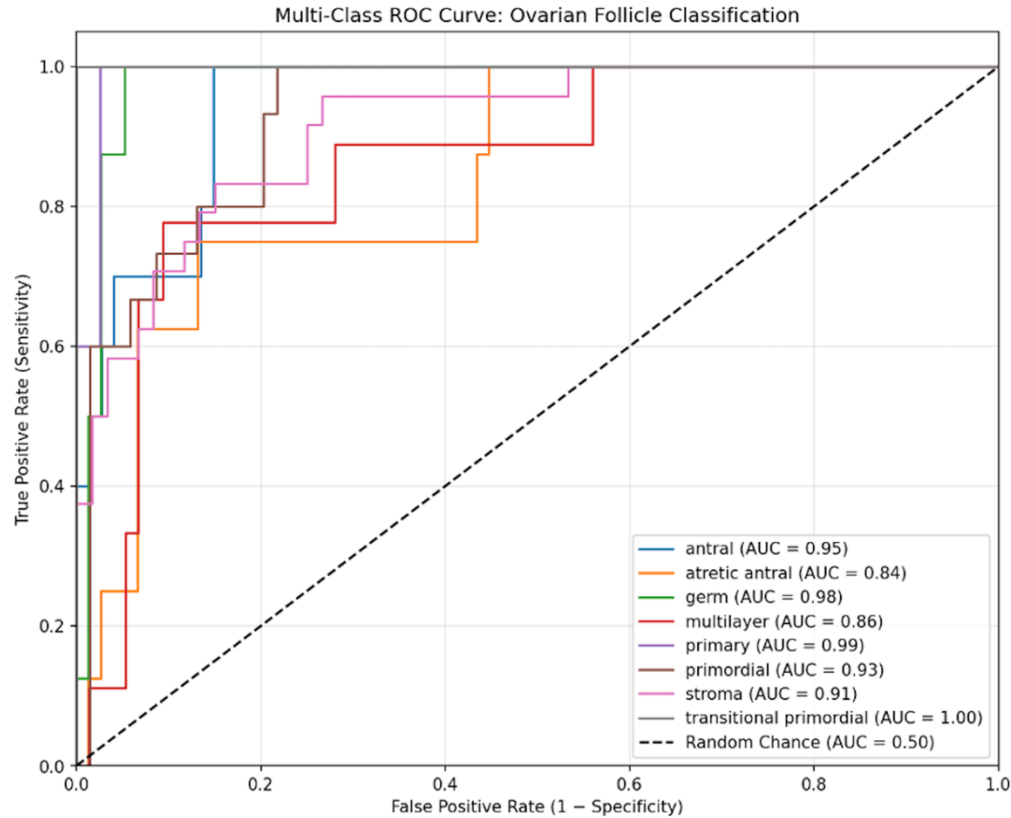


Figure 4. Receiver Operating Characteristic Curve for Ovarian Follicle Classification

Application to Whole Slide Images (WSI)

Once trained, the model was used to analyze the nine whole slide images (WSI) of *H. glaber* histology slides pulled from the MOTHER database. Slides were read in the .tif format using the tiff file Python library with the imagecodecs backend. The distribution of follicle classes can be observed across all nine slides (Figure 5). Each slide was tiled into non-overlapping 224×224 -pixel patches, and a mean-intensity threshold of 220 was applied to exclude background tiles. The number of tissue tiles extracted ranged from 272 to 3,077 (see Table T1 in Appendix), with a strong linear relationship between slide area and tile yield (see Figure A2 in Appendix).

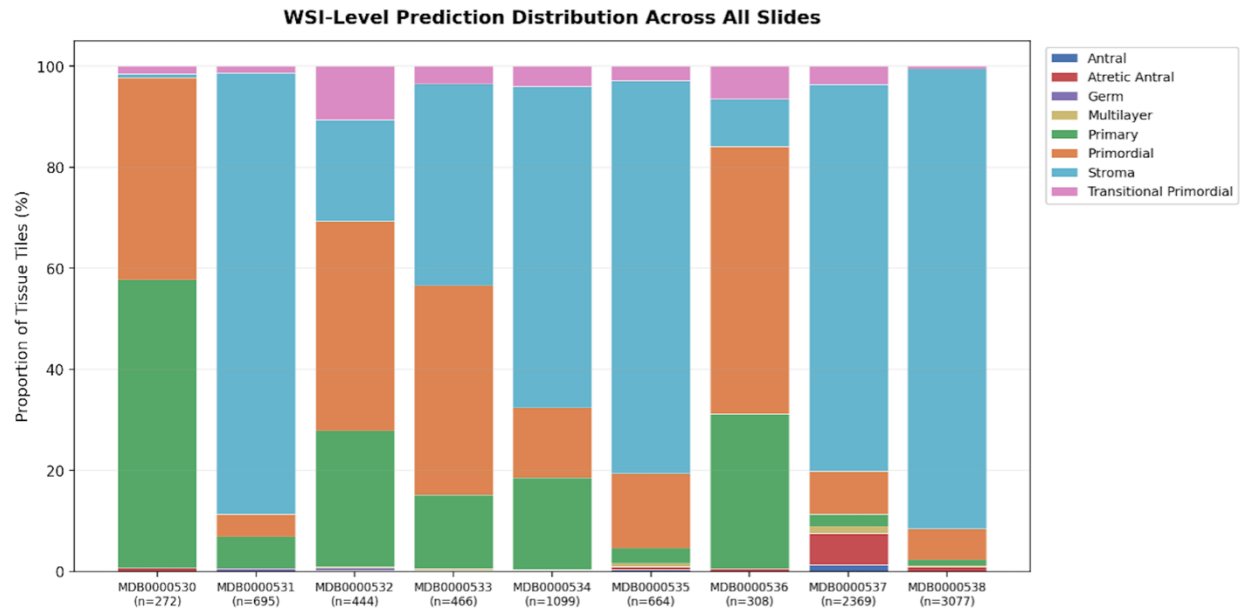


Figure 5. Distribution of WSI-level predictions across the nine *H. glaber* slides pulled from the MOTHER database illustrated in a stacked bar graph.

Class-Specific Patterns Across Slides

The heatmap in Figure 6 reveals variation in class-level predictions across slides. Atretic antral predictions were concentrated in the two adult slides: MDB0000537 (147 tiles) and MDB0000538 (24 tiles), while all other slides had between 0 and 3 atretic antral predictions. Both MDB0000537 and MDB0000538 are from 3-year-old donors, the oldest in the cohort.

Germ cell predictions were near zero across all slides (0-3 tiles per slide), despite germ cells having 42 training tiles. Stroma-to-primordial ratios varied substantially between slides. MDB0000530 and MDB0000536 had very low stroma counts (2 and 29 tiles respectively) with high primary and primordial counts, while MDB0000538 had high stroma counts (2,803 of 3,077 tiles).

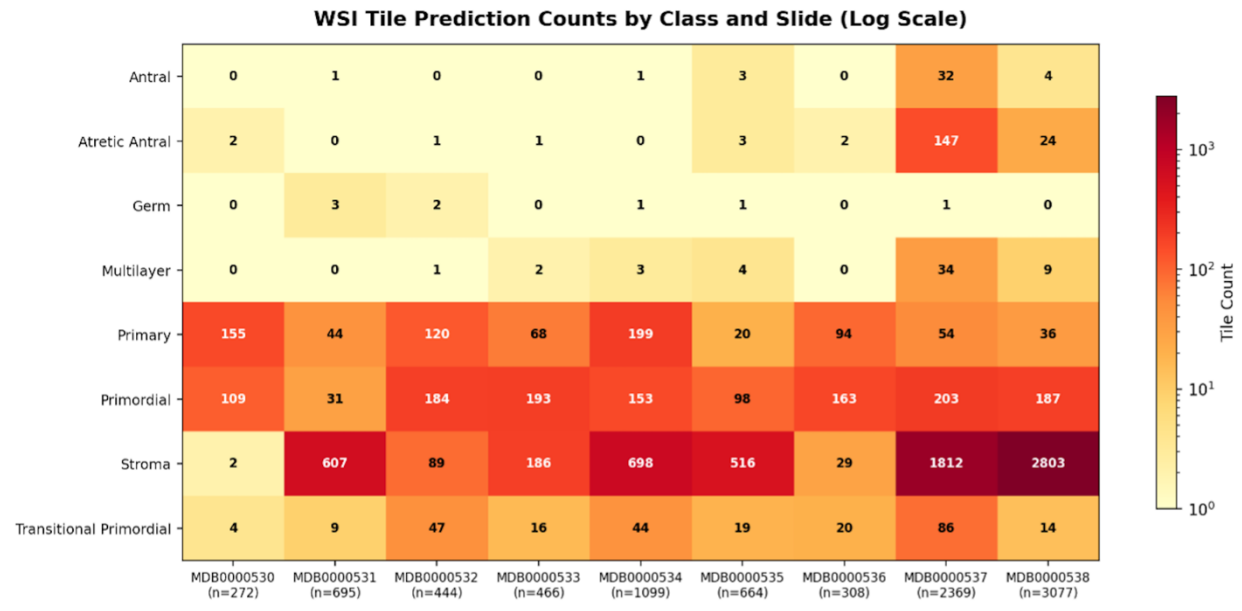


Figure 6. Distribution of WSI tile prediction counts by follicle classes across the nine *H. glaber* slides pulled from the MOTHER database illustrated in a stacked bar graph.

Overall, the ResNet34 classifier demonstrated promising proof-of-concept performance for tile-level classification of ovarian follicle-associated tissue regions in *H. glaber* histological images, despite challenges associated with class imbalance and morphological similarity between classes. The model achieved strong discriminative ability, with a mean AUC of 0.93 across all categories and particularly high performance in transitional primordial, primary, and germ classes. While certain classes such as multilayer and atretic antral exhibited lower classification accuracy, likely due to limited representation. Most importantly, the pipeline successfully scaled to whole-slide image analysis, enabling high-throughput follicle detection across all nine specimens with variable tissue coverage. These results support the feasibility of applying machine learning approaches to automate follicle quantification in this species and provide a foundation for future improvements and large-scale histological studies.

Discussion

This project demonstrated that a machine-learning framework can support tile-level classification of ovarian follicle-associated tissue regions and WSI-level aggregation of prediction patterns in naked mole-rat histology images, while also revealing the limitations of applying this approach to a small, age-diverse dataset. Using whole-slide images from the MOTHER database, XML-derived metadata, QuPath-based annotations, and CNN-based classification, we established a reproducible pipeline for generating tile-level predictions and extending them to whole-slide analysis.

The results provided an initial characterization of follicle-stage distributions across donors of different life stages and indicated that classification performance was stronger for some tissue classes than for others. Overall, these findings support the promise of automated ovarian histology analysis in the naked mole-rat but also emphasize the importance of continued refinement through expanded annotation, larger training sets, and further optimization of the modeling framework.

The performance of the ResNet34 convolutional neural network (CNN) was evaluated across two training phases: feature extraction (frozen) and fine-tuning (unfrozen). During Phase 1 (frozen training), the model achieved its best performance with a validation loss of 0.785 and an error rate of 26%. This indicates that the pretrained convolutional layers were effective at extracting generalizable features from image tiles, allowing the classifier head to distinguish between follicle classes with moderate accuracy. The relatively low validation loss suggests stable learning and reasonable alignment between predicted and true class distributions at this stage.

In Phase 2 (unfrozen training), where all layers were fine-tuned, the optimal model was identified at the first epoch following unfreezing. Subsequent epochs showed a consistent increase in validation loss, triggering the “Early Stopping Callback” to halt training and restore the best-performing weights. This pattern suggests that further fine-tuning led to overfitting, where the model began to specialize excessively to the training data at the expense of generalization. The degradation in validation performance implies that the pretrained features were already near-optimal for this task, and additional weight updates disrupted useful feature discrimination rather than improving them.

The evaluation on the independent test set resulted in a higher error rate of 32%, indicating a drop in performance when applied to unseen data. This gap between validation (26%) and test performance suggests limited generalization, potentially due to factors such as dataset variability, class imbalance, or domain shift between training/validation tiles and test images. It may also reflect the remaining inherent difficulty of distinguishing between morphologically similar follicle classes, particularly in cases where transitional features are subtle, as seen in the confusion matrix. While this performance supports feasibility, it did fall short of the reliability originally expected of about 20-25% [14]. The resulting 32% error highlights dataset and annotation limitations and need for further optimization.

The confusion matrix indicates that the ResNet-34 model performs well on clearly defined follicle classes, with higher accuracy observed for transitional primordial, germ, and primordial categories (Figure 3). Moderate performance was seen in primary and antral follicles, while more complex classes such as stroma, atretic antral, and especially multilayer showed substantial misclassification. The three classes also obtained the lowest AUC values from the ROC curve analysis: stroma (AUC = 0.91), multilayer (AUC = 0.86), and atretic antral (AUC =

0.84). These AUC values indicate that the model retained moderate-to-strong class discrimination for these categories, but the confusion matrix shows that discrete classification accuracy remained limited for multilayer and atretic antral follicles [11]. Overall, most errors occurred between biologically similar or developmentally adjacent classes, again suggesting that while the model captures general morphological features, it struggles with finer distinctions between overlapping follicle stages.

Once the nine whole-slide images (WSI) of *H. glaber* histology slides were tiled to scale, the CNN assigned each tile to one of the eight classes. The classes of antral and atretic antral follicles were predominantly identified on the adult specimen slides, MDB0000537 and MDB0000538. This pattern is seen in rodents and other mammals as adult specimens have a more robust and mature endocrine system in which hormones, such as gonadotropin, support the cycling of ovaries and production of mature follicles [12]. Due to the more active and rapid folliculogenesis observed in adult mammals, the higher occurrence of antral and atretic antral follicles identified in these adult specimens, ranging from 4 tiles to 147 tiles, compared to the remaining neonatal and prepubertal slides, which do not exceed more than 3 identified tiles in either class, is expected. These WSI-level prediction patterns are biologically plausible but should be interpreted as model-generated hypotheses rather than validated follicle counts.

These adult slides were also composed of a high ratio of stromal tiles. The microenvironment provided by stroma is one of the most important aspects of ovarian health and folliculogenesis, as it supports the cells through connective tissue and vasculature. Many studies discuss the extensive role stroma plays in the restructuring and reorganization of tissue that comes with aging and the maturation of ovaries, supporting the results seen within Figure 6 [13]. The neonatal and prepubertal specimens were primarily composed of stroma, primary,

primordial, transitional primordial follicles, all found in early-stage development and folliculogenesis. One notable result was the limited identification of germ cells, specifically in the neonatal slides. This could be attributed to some of the limitations of the CNN and the sophistication of the QuPath annotator.

Recent advances in deep learning have enabled the development of automated pipelines for ovarian follicle detection and classification, with methodological differences often driven by dataset scale and annotation strategy. One approach, described by İnik et al. (2019), utilizes a two-stage framework that separates segmentation and classification, allowing follicles to be explicitly localized before categorization [14]. In contrast, the present MOTHER-based pipeline, developed on a limited dataset of nine *H. glaber* whole-slide images, employs a tile-based ResNet34 classifier that assigns labels to fixed image patches without explicit segmentation. While this approach allows for efficient processing of whole-slide images, follicle quantification remains indirect and dependent on tile-level predictions, which may reduce precision in regions with dense or overlapping structures as compensation for faster processing.

Another distinction lies in the underlying data representation and annotation strategy. The framework proposed by İnik et al. relies on pixel-level or object-level annotations to train segmentation models, providing detailed structural information at the cost of increased manual and computational processing effort [14]. In contrast, the present study utilizes tile-level annotations derived from QuPath, which are more efficient to generate but inherently less granular. Although this strategy supports feasibility within a small dataset context, the limited number of slides may restrict the diversity of morphological patterns captured during training. Consequently, while segmentation-based approaches offer higher spatial precision, the current

method prioritizes practicality and computational efficiency, albeit with reduced anatomical specificity.

These methodological and dataset constraints have important implications for model performance and generalizability. Segmentation-classification pipelines are better suited for precise follicle quantification and detailed morphological analysis, particularly when trained on larger and more diverse datasets. In contrast, the current approach demonstrates that meaningful classification and slide-level pattern detection can still be achieved in a small-cohort setting, including the identification of age-associated variation across samples. However, the limited sample size and tile-based framework may contribute to instability in rare or ambiguous classes and reduce robustness across broader datasets. Taken together, this highlights a key trade-off between precision, scalability, and dataset size in computational histopathology, and suggests that future work should incorporate larger cohorts and potentially hybrid segmentation-classification strategies to improve both accuracy and generalizability. In a future approach, a segmentation model could first detect and delineate individual follicles as discrete objects, separating them from the stroma or overlapping structures. This could then be assigned through a classifier into a developmental stage. These additions to the approach can reduce the reliance on fixed tile boundaries and limit background noise to improve accuracy in complex regions such as those found in adult specimens.

Considering recent approaches for automated histological classification, particularly in complex tissues such as the ovary. A key strength of this study is the strong overall performance of the ResNet34-based pipeline in classifying ovarian follicle subtypes, achieving a mean AUC of 0.93 across classes. The model learned biologically meaningful patterns, with high accuracy in well-represented and morphologically distinct classes such as primordial, germ cells, and

transitional primordial follicles. The use of class weighting, data augmentation, and strict training, validation, and test separation contributed to stable generalization despite class imbalance. More broadly, these results indicate that deep learning can reliably extract consistent histological signals from complex ovarian tissue morphology in *H. glaber*.

A second strength is the successful application of the model to whole-slide image analysis, enabling scalable, high-throughput quantification across all nine specimens. The pipeline handled large variations in tissue coverage (272 to 3,077 tiles per slide) while maintaining coherent class-level predictions. Notably, it also captured biologically plausible variation across developmental stages, including elevated atretic antral predictions in the oldest donors (MDB0000537 and MDB0000538). This suggests that the model is not only classifying tissue features but also reflecting meaningful age-related differences in ovarian structure, highlighting its potential for biological interpretation beyond classification accuracy alone.

However, several limitations remain in both class-level performance and slide-level consistency. Rare or morphologically ambiguous classes, particularly multilayer, atretic antral, and germ cells, showed reduced stability, with germ cells predicted at near-zero levels across slides despite being present in training data. In addition, strong variability in stroma-to-primordial ratios across slides as seen on the heatmap Figure 6, suggests sensitivity to tissue composition and possible batch or sampling effects. These findings indicate limitations in class separability and dataset representativeness. This can broadly suggest that technical and biological variation across slides may influence model outputs in ways that require further normalization or domain adaptation in future work.

Another limitation of the model is evident in the baseline test error rate of 32%. Although pretrained ResNet34 ImageNet weights offer a strong starting point, they are optimized for natural images rather than ovarian histology, which can contribute to a gap between validation and test performance. This could be improved through deeper fine-tuning of earlier network layers with differential learning rates, so the model better learns tissue-specific features such as follicle boundaries and stromal texture [15]. Histology-specific augmentation, including rotations, flips, stain jittering, and scale variation, would improve robustness to slide variability, while stain normalization methods such as Macenko or Reinhard could reduce differences between validation and test images by standardizing stain color distributions [16, 17]. Together, these strategies would better adapt ResNet34 to histology data and help narrow the validation-to-test performance gap.

This application of a machine-learning framework for ovarian follicle classification and quantification on a recently added species to the MOTHER database fills a gap in current research for histological analysis and automation of reproductive tissues in mammals. Our hypothesis that machine learning could automate time-intensive tasks in follicle identification and counting was validated in model evaluation results. However, the current performance should be interpreted as a proof of concept rather than a fully optimal solution, as further improvements in data scale, and annotation quality is required to achieve higher reliability.

The targeted focus of our research leaves the cross-species application of a machine learning pipeline unanswered, creating a possible direction for future work. Using QuPath for tile creation prior to training the model, creates a need for a similar step to be done before applying the model to another species in the MOTHER database. Additionally, the limited WSI available in the MOTHER database for *H. glaber*, creates a need for future work to include more training and

validation slides for consistency and efficacy of the model. This could improve post-training model evaluation results, leading to more significant findings. It may also impact the ability for cross-species application, if the increase in sample size can allow for more specific training and validation. This research is significant as it contributes to the growing use of machine-learning in reproductive research, more specifically in histological image analysis for follicle identification and classification.

Conclusion

This study demonstrates how integrating machine learning with histological image analysis can support more reproducible analysis of ovarian tissue. By combining curated slide data from the MOTHER dataset with QuPath-derived annotations, we developed a pipeline that extends traditional manual assessment toward automated, reproducible image analysis. The use of a ResNet34 CNN classifier enabled tile-level prediction of multiple follicle-associated tissue classes across whole-slide images, addressing a common bottleneck in histological workflows.

Equally important, the exploratory data analysis provided critical insight into slide variability, staining differences, and spatial heterogeneity. All of these qualities are factors that directly influence model performance. These findings informed preprocessing decisions, including tissue masking and tile selection, which reduced background noise and helped ensure that the model learned from biologically relevant tissue regions. They also indicate that future iterations should evaluate stain normalization methods to reduce stain-related bias across slides. Together, these preprocessing choices support more consistent feature extraction and may improve generalization to unseen images by limiting the influence of background regions and

slide-specific artifacts. This highlights the importance of integrating exploratory data analysis directly into model development rather than treating it as a separate preliminary step.

While the results highlight the promise of ML approaches in ovarian histology, they also underscore ongoing challenges, including variability in staining, annotation quality, and the need for larger, more diverse datasets. Future work can build on this foundation by expanding the dataset, incorporating more elaborate segmentation-based models, and extending analyses beyond basic histology structure to elements like tumor microenvironment. Ultimately, this project supports a broader shift toward scalable, quantitative histopathology, where ML-assisted tools enhance both research and clinical insight into ovarian biology.

Data and Code Availability

Code Development: Debugging support and logic optimization for data parsing, image analysis, and model training pipelines were facilitated by ChatGPT (GPT 5.2 Thinking) and Claude (Opus 4.6).

The data underlying this study are available on ReproAnalytics' GitHub repository:

<https://github.com/ReproAnalytics>

Conflicts of Interest

The authors declare no conflicts of interest.

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